

# MOHAMMED LOKHANDWALA

Surat, Gujarat

+91-7486820413    lokhandwalamohammed100@gmail.com    Mohammed    Mohking1    Portfolio

## EDUCATION

### P.P. Savani University

B.Tech in Computer Science (AI & ML specialization) — CGPA: 8.1

Sep 2022 – Sep 2026

Surat, Gujarat

## EXPERIENCE

### Avinashi Group of Companies

Aug 2025 – Jan 2026

#### AI Engineer Intern

Surat, Gujarat

- Built a **sub-250ms bidirectional telephony AI agent** for live PSTN calls, using **20ms audio micro-batches** with streaming STT-LLM-TTS, WebRTC/WebSockets, noise handling and barge-in interruption control.
- Designed a **document retrieval pipeline** for dense business documents using hierarchical semantic chunking, **hybrid vector/lexical retrieval**, reranking and metadata-aware context assembly.
- Built a **financial document processing pipeline** for scanned statements, invoices and receipts, combining layout/table detection, OCR, custom preprocessing and post-processing to produce structured data.
- Designed shared **AI infrastructure** around Supabase and a custom API gateway with authentication, organization-level access, quotas, storage and reusable endpoints for multiple AI services.

### Seepossible Innovative Solutions LLP

Jun 2024 – Jun 2024

#### AI/ML Intern

Surat, Gujarat

- Built an internal **image upscaling API** with authentication and configurable model, resolution and size parameters for use as a reusable AI microservice.
- Adapted a **frame interpolation workflow** for jewelry rendering, allowing roughly half the frames to be rendered directly and cutting rendering/post-processing time by about **50%**.

### Demonictl / Community Translation Platform

Collaborative Project

#### Web Platform / Operations

Remote

- Helped build and operate a public **novel-translation platform** that accumulated **500K+ documented views** across published titles, with advertising and reader-supported funding.

## SELECTED PROJECTS

### Generalized RL Locomotion Platform (Python, PyTorch, PPO/SAC, Box2D, Gymnasium, DearPyGui)

- Built a modular 2D reinforcement-learning environment where users can define articulated creatures with custom bodies, joints, limits, sensors and colliders, then train them to move through procedural terrain.
- Implemented the physics/control loop, LiDAR observations, dense reward shaping, parallel training, checkpoint management and an interactive rigging/evaluation GUI, with support for PPO and SAC.

### Local ML Experiment Tracker (Python, Node.js, React, TypeScript, SQLite, MongoDB)

- Built an offline ML experiment tracker from scratch for logging runs, metrics, configurations and artifacts, with a Python SDK, CLI and web dashboard.
- Implemented experiment comparison, metric aggregation, local storage, hardware telemetry and integrations for common ML training workflows.

### AI Video Ad Generator (Python, LLMs, TTS, Lip-sync, Video, FFmpeg)

- Built a dual-mode video generation system combining a conversational agent with predefined multi-scene pipelines for creating product ads from natural-language inputs.
- Coordinated image, video, voice, lip-sync and subtitle generation with validation, retries, provider fallbacks and FFmpeg-based composition across multiple scenes.

### MailDoc - Multi-Agent Business Automation (Python, MCP, RAG, Mem0, AutoGen, Docker)

- Built an MCP-based workflow that connects generic IMAP email, attachments and documents with retrieval, memory and multi-agent reasoning for business information tasks.
- Added structured tool interfaces and document retrieval so an agent can gather evidence from email and files, answer questions with references and support approval-based workflows.

## TECHNICAL SKILLS

**Languages:** Python, C++, JavaScript/TypeScript, SQL

**AI/ML:** PyTorch, TensorFlow, Computer Vision, NLP, Transformers, RAG, LLMs, Agents, Reinforcement Learning

**Backend/Systems:** FastAPI, Node.js, REST APIs, WebSockets, MongoDB, SQLite, Docker, Linux

**Tools:** Git, GitHub, Vite, W&B

## ACHIEVEMENTS

**GATE:** Cleared GATE in 3rd and 4th year of B.Tech.    **SSIP:** Awarded ₹30,000 for an AI/robotics innovation project.

**Hackathon:** 2nd place in college hackathon.    **Open Source:** Contributed a scraping-method fix to LNReader; change merged upstream.